UNIVERSITI TEKNOLOGI MARA
FINAL EXAMINATION

COURSE	DATABASE DESIGN AND DEVELOPMENT
COURSE CODE	ICT450
EXAMINATION	NOVEMBER 2026
TIME	3 HOURS

INSTRUCTIONS TO CANDIDATES

1. This question paper consists of five (5) questions.
2. Answer ALL questions in the Answer Booklet. Start each answer on a new page.
3. Do not bring any material into the examination room unless permission is given by the invigilator.
4. Please check to make sure that this examination pack consists of:
   i) the Question Paper
   ii) an Answer Booklet - provided by the Faculty
5. Answer ALL questions in English.

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO

This examination paper consists of 6 printed pages.

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QUESTION 1 (20 marks)

a) File system method is an improvement over manual system. However, there are many problems that associated with file system including data anomalies.

i) Briefly describe TWO (2) types of data anomalies.

(4 marks)

ii) Briefly describe TWO (2) applications of big data implemented in healthcare industry. Give an example for each application.

(4 marks)

b) Data redundancy is common issue in file system. Illustrate with description how data redundancy could occur.

(8 marks)

c) Based on the Entity Relationship Diagram in Figure 1.0, explain how referential integrity are enforced.

(4 marks)

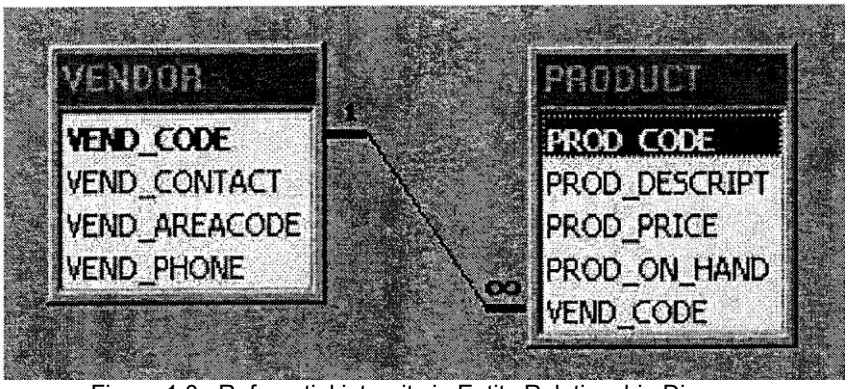


Figure 1.0 : Referential integrity in Entity Relationship Diagram

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QUESTION 2 **(20 marks)**

a) A relational data model is one of the data models that have been used in a database environment. Describe TWO (2) basic features of the relational data model.

(4 marks)

b) A database constraint is a rule that maintains data integrity. Briefly explain TWO (2) examples of domain constraints that can be placed on the data.

(4 marks)

c) Database normalization is a vital process in relational database design that helps minimize data redundancy and improve data integrity. Briefly explain the THREE (3) main types of anomalies that can occur in a denormalized table.

(6 marks)

d) Describe THREE (3) primary challenges associated with mobile databases. Provide examples to support your answer.

(6 marks)

QUESTION 3 (20 marks)

The following questions will be based on data shown in Table 1.0, which was imported from an Excel spreadsheet. This represents a college course information that are offered. Each course is stored in a separate file. The management of data is complex and creates data redundancy. Therefore, to avoid from data redundancy the structure needs to be normalized.

Table 1.0 : Sample of college course information

Course Code	ICT450	Course Name	Database Design and Development	
Tutor ID	306612	Tutor Name	Professor Albus Dumbledore	
Student ID	Student Name	DOB	Gender	Last Attendance Date
2016123456	Luke Skywalker	11.10.96	F	31.01.18
2016534363	Kylo Ren	12.12.96	F	15.01.18
2016788924	Thor Loki	27.06.96	F	31.01.18
2016329874	Clint Barton	31.08.96	F	17.01.18
2016324793	General Hux	23.05.96	F	23.01.18
2016479625	Leia Organa	17.02.96	M	15.01.18
2016328796	Hermione Granger	15.04.96	M	31.01.18

a) State the normal form of the above Table 1.0. Explain your answer with reasons.

(3 marks)

b) Given that information, draw its dependency diagram and identify all transitive and partial dependencies.

(7 marks)

c) Normalize the table until third normal form (3NF). Show all your work.

(10 marks)

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QUESTION 4 **(20 marks)**

The following describes a relational schema for football teams in Malaysia:

```
TEAM (TEAM_ID, TEAM_NAME)

GAME (GAME_ID, GAME_NAME)
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FOOTBALL (FOOTBALL_ID, TEAM_ID*, GAME_ID*, DATE, TIME, LOCATION) PLAYER (PLAYER_ID, P_NAME, ADDRESS, PHONE_NO, SALARY, POSITION, TEAM_ID*)
Write SQL commands to :

a) Create the PLAYER table with integrity requirements.

(3 marks)

b) Insert a row of data, as follows, into the PLAYER table.
PLAYER_ID: 10067
P_NAME: Safiq Rahim
ADDRESS: No.168, Damai Country Height, 40100, Shah Alam, Selangor
PHONE_NO: 019-3451212
SALARY: RM30,000.00
POSITION: Defence
TEAM_ID: FAM326

(2 marks)

c) All strikers for The Red Warrior team are given $10%$ salary increment. Modify their salaries.

(2 marks)

d) Show all the teams' details who played at Bukit Jalil National Stadium in May 2018.

(3 marks)

e) Calculate how many games are played in the first quarter of 2018.

(3 marks)

f) Calculate how many games are played for by each team.

(3 marks)

g) List down the teams' names and players' names who played at Merdeka Stadium and whose salary is RM2,000.00 or RM3,000.00.

(4 marks)

QUESTION 5 (20 marks)

Use the following business rules to create the ERD for a consulting company named BestEverConsult.

- A client can have contract on many projects, but one client must be responsible for contract for each project
- A project may have many employees assigned to it, and each employee may be assigned to many projects.
- Some employees are not assigned to any project
- One employee leads each project
- An employee may lead more than one project at a time
- One employee coordinates each project
- An employee may coordinate more than one project at a time
- Some employees never lead or coordinate projects
- Many of the consulting company employees have multiple job skills (For example, an employee may be qualified as a systems analyst and as a network administrator)
- Each job description – systems analyst, database designer, accountant and so on – can fit many employees (For example the consulting company is likely to have several database designers, systems analysts, etc)
- All work done by the employees for any project is charged to that project.

Based on the given information:

a) Construct a complete Entity Relationship Diagram (ERD) with all required entities, attributes, relationships as well as the cardinalities using the Crow's Foot notation that reflects the given business rules.

(16 marks)

b) Briefly describe TWO (2) examples of report that can be generated from ERD in (a).

(4 marks)

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